Aiden Fox

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Education

University of Southern California — Los Angeles, CA Aug 2024 – Dec 2027

BS Computer Engineering & Computer Science, GPA 3.61

Relevant coursework: Embedded Systems, Digital Circuits, Linear Circuits, IoT, Data Structures, Algorithms, Distributed Systems. Stanford University summer session, CompE & CS, GPA 3.66.

Experience

ML / Fullstack Software Engineer Intern, Motif Inc. — Los Angeles, CA Jun 2026 – Present

- Diagnosed and fixed a message-redelivery defect in a production async worker, where a ten-message SQS batch lease heartbeat only the in-flight message and silently redelivered the rest.
- Ran a security and boundary review of the workflow platform, surfacing a sandbox output DoS, a missing DAG node cap, and a 500-instead-of-409 create race.

Software Engineer, Contract, Energy Connect — Remote May 2026 – Present

- Diagnosed a decimal-truncation defect spanning five layers, from schema validation through to a capture regex, across a multi-tenant logistics platform.
- Closed 600+ tracked issues across four customer portals in four months, including 30 marked urgent.

ML / Fullstack Software Engineer Intern, Motif Inc. — Los Angeles, CA May – Dec 2025

- Cut batch-processing time roughly 65% by scaling an OCR-routed SageMaker captioning pipeline across 22,000+ images with multi-GPU EC2 sharding.
- Engineered fault-tolerant async processing on AWS, refactoring a Voice Comment backend onto Lambda and SQS with retries and CloudWatch observability.

Product Engineering Intern, Advanced Image Robotics — San Diego, CA Nov 2023 – May 2024

- Shipped 26 production AIR One cameras with embedded multi-axis control and stability verification, and integrated a Jetson Nano animation system driving startup sequencing and camera UX.

Projects

Fennec Autonomous Quadruped Teensy 4.1, micro-ROS, ROS 2 Humble, LIDAR-Inertial SLAM

- Ran a servo system-identification campaign on a custom 2.5 kHz logger, isolating a 75 ms command-to-motion deadtime as friction rather than undervoltage by re-testing across supply rails, and cross-validating peak speed against datasheet to within 1.5%.
- Overturned a first-order actuator lag model that would have corrupted sim-to-real transfer, replacing it with a rate-limiter, delay and deadband model after loaded captures showed speed was governed, not torque-limited.
- Designed the custom 4-layer power distribution board in KiCAD (buck regulators, INA226 current telemetry, MOSFET reverse-polarity chain), then proved its mounting-hole keepouts effective rather than merely specified by testing 196 sampled points against 45 filled copper polygons, surfacing an unrecorded washer-overhang short path across the battery-negative standoff.
- Built a Feetech STS3215 half-duplex bus driver from scratch (PING, READ_DATA, WRITE_DATA, SYNC_WRITE) with no vendor SDK dependency, covered by frame-encoding and bus-timing unit tests, on a 200 Hz IntervalTimer ISR control loop with a software watchdog.
- Patched a double queue-drain defect in the upstream Unitree L2 IMU bridge, restoring the sensor's 250 Hz stream; released as a public fork and used for SLAM bring-up.

FPGA Snake Game Verilog, Digital Design, Xilinx FPGA

- Implemented real-time Snake fully in hardware across six Verilog submodules (game logic, VGA driver, SPI master, clock generation, seven-segment display, top-level), with accelerometer-over-SPI tilt control and an 8-bit LFSR for fruit placement.

Technical Skills

Languages: C, C++, Python, Verilog, Java, TypeScript

Embedded & Hardware: Teensy 4.x (Cortex-M7), Raspberry Pi, Jetson, FPGA (Xilinx, Vivado), RTOS, KiCAD, PCB bring-up, digital logic, I2C, SPI, UART, ADC, logic analyzers

Real-Time & Systems: ISR control loops, jitter characterization, watchdogs and safety FSMs, ROS 2, micro-ROS, Linux, sockets, AWS, Docker, Git